

DATA ANALYSIS WITH PYTHON

Student Career Transformation Program

Program Overview

This intensive 12-week program is designed to bridge the gap between classroom theory and industry practice. Our mission is to equip students with the technical proficiency and analytical mindset required to succeed in the global data economy. We focus on hands-on learning, portfolio building, and job-readiness.

Detailed Curriculum

Module	Learning Focus	Key Outcome
Week 1-2: Setup & Syntax	Python Fundamentals, Lists, Loops, and Functions.	Writing clean, logical Python code scripts.
Week 3-5: Data Wrangling	NumPy and Pandas deep-dive; handling datasets.	Cleaning and preparing "real-world" messy data.
Week 6-8: Visualization	Storytelling with Matplotlib and Seaborn.	Building professional dashboards and reports.
Week 9-10: Database Access	SQL basics and connecting Python to SQL.	Querying and extracting data like a pro.
Week 11-12: The Capstone	Portfolio project development and GitHub setup.	A live project for your LinkedIn and Resume.

Hands-On Projects

Beginner

Project 1: Global Health & Wealth Trends

Using the Gapminder dataset to analyze life expectancy and income across 100+ nations. Students learn to handle multi-dimensional data and create animated trend charts.

Intermediate

Project 2: E-commerce Customer Behavior

Analyzing retail transaction data to identify purchasing patterns, high-value customers, and seasonal sales peaks. Focuses on Pandas GroupBy and Pivot Tables.

Career & Professional Development

- ✓ **Resume Workshop:** Integrating technical keywords.
- ✓ **GitHub Profile:** Showcasing clean, documented code.
- ✓ **LinkedIn Strategy:** Networking for Data roles.
- ✓ **Mock Interviews:** Solving live data challenges.